Event-Aware Market Making in Prediction Markets

Anticipated Information and Stale-Quote Execution

Model specification

October 14, 2026

Abstract

We specify an inventory-sensitive market maker for a binary prediction contract. Fair value evolves through continuous diffusion, recurring unscheduled news, and a single anticipated information event with a time-dependent arrival hazard. Information shocks may execute against outstanding quotes before cancellation, so news affects both inventory risk and the immediate economics of quoting. The control problem reduces to two coupled Hamilton–Jacobi–Bellman partial integro-differential equations, corresponding to the anticipated event being pending or completed. We give the event operator, terminal conditions, a qualified spread-widening result, and a reproducible Monte Carlo comparison with a constrained diffusion benchmark. The numerical experiment measures performance under synthetic news and execution assumptions. We then replace the single two-point news kernel by a mixture of two-point kernels, which keeps the event operator mean-preserving while allowing heavy-tailed jumps, and admit endpoint quotes as an explicit withdrawal control. Applied to Kalshi's New York daily-high-temperature contracts and replayed against the public trade tape, the mixture-kernel controller, calibrated on October 2025–March 2026, earns \$11.03 a day out of sample (April–September 2026) against \$5.27 for the two-point kernel and \$3.31 for the diffusion benchmark, and learns to withdraw around scheduled station reports without a hand-coded calendar.

1 Scope and modeling conventions

This document formalizes the event-aware market-making framework developed in the shared research discussion. Its defining change is to distinguish *when information arrives* from *how much uncertainty it resolves*, and to allow the resulting price jump to trade against a stale quote. The anticipated event occurs at most once; unscheduled news can recur.

The baseline uses certain execution of a crossed stale quote, a fixed trade size, and no fees. Quotes are predictable: the maker chooses them before observing the information shock. After the shock and any stale execution, the maker may reprice. Ordinary order arrivals convey no additional information about the contract payoff in this specification.

We retain the source model's equations while making the following conventions explicit: both news kernels preserve conditional mean; prices lie in $[0, 1]$; running event risk uses pre-event inventory; and unconditional variance matching includes the probability that the one-shot event is still pending. Numerical choices and empirical hypotheses below are distinguished from model identities.

2 Prior work and relationship to this model

2.1 Direct foundation: prediction-market optimal control

Feil and Nendel [1] provide the direct foundation: a bounded martingale belief generated by a latent diffusion, fixed-size executions, inventory-sensitive quoting, and running and terminal risk penalties. Their Sections 2–3 develop the control problem and its reduced Bellman equation; Example 2.1 supplies the logistic diffusion behind the factor $p(1-p)$ used here. The decomposition $U = x+qp+V$ retains that structure.

There are material differences. Their model has continuous pre-resolution beliefs, a finite inventory grid, general time- and belief-dependent execution intensities, and quotes in $[0,1]^2$. Our straddling constraint $b \leq p \leq a$ and symmetric exponential intensities are additional restrictions. Their payoff is revealed immediately after the trading horizon, explaining the residual settlement penalty. Their classical-solution and verification results do not automatically extend to our nonlocal, regime-switching equations with atomic kernels and discontinuous fill indicators.

2.2 Broader inventory and adverse-selection literature

Avellaneda and Stoikov [2] establish a canonical inventory-based quoting framework with diffusive prices and quote-dependent Poisson executions, including exponential arrival intensities. This is the broader lineage of our ordinary-fill Hamiltonian. Their exponential-utility criterion differs from the additive inventory penalties used here; their closed-form approximations should not be imported unchanged into this model.

Cartea, Jaimungal, and Penalva [3], particularly Chapter 10 and Section 10.4, provide broader context for stochastic-control market making and adverse selection. We cite this as methodological background, not as the source of our particular one-shot event operator. The present model represents adverse selection through a specified news-triggered crossing rule rather than estimating a full model of informed order flow.

2.3 Empirical motivation for anticipated information

Frino and Hill [4] study scheduled macroeconomic announcements in Australian futures markets. They report significant spread widening during the 20 seconds before announcements and the 30 seconds afterward. This supports investigating time-localized quoting responses, but does not establish our prediction-market calibration or certain-pickoff assumption.

Siikanen, Kanniainen, and Valli [5] distinguish scheduled and unscheduled company announcements using NASDAQ Nordic order books. They find announcement-related changes in liquidity and emphasize that multi-level depth can behave differently from the best bid–ask spread; scheduled announcements can improve multi-level liquidity. Their findings motivate separate event types and caution against interpreting wider spreads as a complete measure of liquidity deterioration. Neither empirical study proves this model's comparative statics or measures its proposed economic advantage.

2.4 Model extensions

Relative to the direct diffusion foundation, this model adds recurring unscheduled news, a one-shot anticipated event with survival-dependent arrival, and execution at pre-event quotes after fair value changes. The risk criterion extends the diffusion variance charge to predictable quadratic

variation. These additions yield the coupled equations below and make event exposure enter quote optimization directly through $\mathcal{G}^K_{b,a}$.

The event operator depends on the controls and changes inventory, unlike a quote-independent jump operator. The pending/completed state prevents repeat arrivals of the same anticipated release. These features describe the model's scope; they are not claims of priority over the entire market-microstructure literature. The two-controller numerical comparison below measures the combined effect of these extensions; isolating timing information remains a further test.

3 State, fair value, and information arrivals

3.1 Contract and state variables

The contract pays $Y \in \{0,1\}$ at resolution. Over a trading horizon $[0,T]$, define the fair probability by

$$p_t = \mathbb{E}[Y \mid \mathcal{F}_t] \in [0,1]. \tag{1}$$

The maker holds cash X_t and inventory $Q_t \in \Delta\mathbb{Z}$, where $\Delta > 0$ is the quantity executed per fill. Let

$$S_t = \mathbf{1}_{\{\tau \le t\}} \in \{0,1\} \tag{2}$$

indicate whether the anticipated information event at time τ has occurred. The Markov state is (t, X_t, p_t, Q_t, S_t). The horizon may precede payoff resolution; conditional expected terminal inventory value is then $Q_T p_T$. If T is resolution itself, $p_T = Y$ and residual Bernoulli variance vanishes.

3.2 Continuous variation and unscheduled news

Between information events,

$$\mathrm{d}p_t = \sigma(t,p_t)\mathrm{d}W_t, \qquad \sigma(t,p) = \bar{\sigma}(t,p)p(1-p). \tag{3}$$

Assume coefficients are regular enough for a well-defined process that stays in $[0,1]$. Unscheduled shocks arrive with intensity $\lambda_u(t,p)$. At a shock, the new probability has kernel $K_u(\mathrm{d}p' \mid p)$ supported on $[0,1]$, with

$$\int p' K_u(\mathrm{d}p' \mid p) = p, \qquad v_u(p) = \int (p'-p)^2 K_u(\mathrm{d}p' \mid p). \tag{4}$$

Thus neither diffusion nor unscheduled news creates predictable fair-value drift. Kernels may be extended to depend on time, with corresponding changes to notation.

3.3 The anticipated event

Let F be the arrival-time distribution, with density f on the modeled window. Conditional on survival, its hazard is

$$h(t) = \frac{f(t)}{1-F(t)}, \qquad \mathbb{P}(\tau \in [t, t+\mathrm{d}t] \mid \tau > t) = h(t)\mathrm{d}t + o(\mathrm{d}t). \tag{5}$$

The actual event intensity is $(1-S_{t-})h(t)$. At arrival, S changes from 0 to 1 and never changes back. The probability transition kernel K_s satisfies

$$\int p' K_s(\mathrm{d}p' \mid p) = p, \qquad v_s(p) = \int (p'-p)^2 K_s(\mathrm{d}p' \mid p). \tag{6}$$

The subscript on K_s denotes the scheduled/anticipated event type, whereas the index on V_s below denotes the regime $s \in \{0, 1\}$. We assume the time-only hazard is valid conditional on the full pre-event state, so that the absence of the event does not itself require a drift update to p. An event time informative about Y would require a richer filtering model.

A finite hazard window generally leaves positive probability that the event has not occurred by T:

$$\mathbb{P}(\tau > T \mid \tau > t) = \exp\left(-\int_t^T h(u)\mathrm{d}u\right). \tag{7}$$

An event guaranteed by a finite deadline instead requires a divergent integrated hazard or an atom, treated as a discrete event condition.

3.4 A mean-preserving two-point kernel

For either event type, a convenient choice is K_ρ, $0 \le \rho \le 1$:

$$p' = \begin{cases} p^+ = p + \rho(1-p), & \text{with probability } p, \\ p^- = (1-\rho)p, & \text{with probability } 1-p. \end{cases} \tag{8}$$

Direct calculation gives

$$\mathbb{E}[p' \mid p] = p, \qquad \mathbb{E}[(p'-p)^2 \mid p] = \rho^2 p(1-p). \tag{9}$$

Use separate parameters ρ_u and ρ_s for the two event types. $\rho = 0$ produces no news and $\rho = 1$ fully resolves the probability. Mean-preserving kernels on $[0, 1]$ are degenerate at $p = 0$ and $p = 1$.

3.5 Mixtures of two-point kernels

A single K_ρ moves the probability by the same relative amount at every arrival. Observed information events are rarely like that: most move the price very little and a few move it a great deal (Section 12). Matching only the variance $\rho^2 p(1-p)$ then places the whole jump at its root-mean-square size, which is larger than the typical move and smaller than the tail. For weights $w_j \ge 0$ with $\sum_j w_j = 1$ and sizes $0 \le \rho_1 < \cdots < \rho_J \le 1$, define

$$K_{\mathbf{w}} = \sum_{j=1}^{J} w_j K_{\rho_j}. \tag{10}$$

Each component is mean-preserving and supported on $[0, 1]$, so the mixture is too:

$$\mathbb{E}[p' \mid p] = p, \qquad \mathbb{E}[(p'-p)^2 \mid p] = \Big(\sum_j w_j \rho_j^2\Big) p(1-p). \tag{11}$$

The event operator is linear in the kernel,

$$\mathcal{G}_{b,a}^{K_{\mathbf{w}}}[W] = \sum_j w_j \, \mathcal{G}_{b,a}^{K_{\rho_j}}[W], \tag{12}$$

so the Bellman equations below are unchanged in form: an event type with intensity λ and kernel $K_{\mathbf{w}}$ contributes the components $(\lambda w_j, K_{\rho_j})$ as separate atomic-kernel terms. The two-point kernel is the case $J = 1$. The mixture keeps the strict-crossing thresholds explicit: on the ask side they are $\rho_j(1-p)$ for each j, on the bid side $\rho_j p$.

Weights can be estimated from observed event windows. For a pre-event probability p_0 and post-event probability p_1, the two-point size that would produce the move is

$$\hat{\rho} = \begin{cases} (p_1 - p_0)/(1 - p_0), & p_1 > p_0, \\ (p_0 - p_1)/p_0, & p_1 \le p_0, \end{cases} \tag{13}$$

and w_j is the share of windows whose $\hat{\rho}$ is nearest ρ_j. Because each $\hat{\rho}$ is assigned to a whole component, the fitted mixture does not in general reproduce the sample variance exactly; report both.

4 Quotes and execution

4.1 Admissible controls

At state (t, p, q, s), the maker posts bid b and ask a in an admissible set $\mathcal{A}(p, q)$. A baseline price constraint is

$$0 \le b \le p \le a \le 1. \tag{14}$$

Define quote distances $\delta_b = p - b$ and $\delta_a = a - p$. If the experiment requires continuous meaningful participation, impose a maximum quote displacement D:

$$0 \le \delta_b \le \min(D, p), \qquad 0 \le \delta_a \le \min(D, 1 - p). \tag{15}$$

Otherwise endpoint quotes can serve as effective protection against pickoff. The restriction that quotes straddle p is an assumption of this model and limits how aggressively the maker can skew to reduce inventory.

Withdrawal as a control. The endpoint quotes $b = 0$ and $a = 1$ ($\delta_b = p$, $\delta_a = 1 - p$) receive, by strict crossing, no stale fills, and ordinary fills at rate Ae^{-kp} or $Ae^{-k(1-p)}$, which is negligible at calibrated k away from the boundaries. In the replay below an endpoint quote is implemented as posting nothing on that side; the solver keeps the small exponential rate, a slight inconsistency we retain for simplicity. When the experiment does not require continuous participation, we drop the cap (15) for these two actions only and add them to the candidate set, so that withdrawal is chosen by the optimization rather than imposed by a calendar. Ordinary quotes remain subject to D.

The analytical state space is $\Delta\mathbb{Z}$. For finite-grid computation, choose inventory limits and explicitly disable any side that would violate them. A disabled side has neither ordinary fills nor stale fills. All formulas below are for active sides; the same availability mask must be applied to both execution mechanisms at inventory boundaries. This is an implementation convention, not an additional source-model assumption.

4.2 Ordinary market orders

Away from news events, ordinary fills have intensities

$$\Lambda^b(\delta_b) = Ae^{-k\delta_b}, \qquad \Lambda^a(\delta_a) = Ae^{-k\delta_a}, \qquad A, k > 0. \tag{16}$$

More general side-specific intensities can replace these functions. The exponential specification follows the classical quoting framework of [2]; the prediction-market foundation in [1] permits more

general intensities. A bid fill sends (x, q) to $(x - \Delta b, q + \Delta)$; an ask fill sends it to $(x + \Delta a, q - \Delta)$. For a reduced value function V_s, write

$$\begin{aligned}\mathcal{F}_{b,a}[V_s](t,p,q) = \Lambda^b(p-b)\big[\Delta(p-b) + V_s(t,p,q+\Delta) - V_s(t,p,q)\big] \\ + \Lambda^a(a-p)\big[\Delta(a-p) + V_s(t,p,q-\Delta) - V_s(t,p,q)\big].\end{aligned} \tag{17}$$

4.3 Stale-quote execution at information events

An information event first reveals p'. Before cancellation, a fast trader executes one lot against any strictly crossed stale quote:

$$I_b(p'; b) = \mathbf{1}_{\{p' < b\}}, \qquad I_a(p'; a) = \mathbf{1}_{\{p' > a\}}, \qquad q' = q + \Delta I_b - \Delta I_a. \tag{18}$$

Equality does not trigger a stale fill. The two indicators cannot both be one under (14). The associated cash is $x' = x - \Delta b I_b + \Delta a I_a$. Define the event operator on a post-event reduced value W by

$$\begin{aligned}\mathcal{G}^K_{b,a}[W](t,p,q) = \int \Big[W(t,p',q') + \Delta(p'-b)I_b(p';b) \\ + \Delta(a-p')I_a(p';a)\Big] K(\mathrm{d}p' \mid p).\end{aligned} \tag{19}$$

Both execution markouts are nonpositive. The operator jointly updates fair value, inventory, and the loss from stale execution.

For clarity, the total mark-to-market change at the event is

$$x' + q'p' - (x + qp) = q(p' - p) + \Delta(p' - b)I_b + \Delta(a - p')I_a. \tag{20}$$

The first term has conditional mean zero because K is mean-preserving; the remaining terms explain (19). This calculation accounts for the simultaneous jump and fill without omitting their interaction.

5 Risk criterion and cash reduction

With no predictable atoms in event time, the predictable quadratic variation is

$$\mathrm{d}\langle p\rangle_t = c_{S_{t-}}(t, p_{t-})\mathrm{d}t, \qquad c_s(t,p) = \sigma^2(t,p) + \lambda_u(t,p)v_u(p) + (1-s)h(t)v_s(p). \tag{21}$$

For $\gamma, \gamma_T \geq 0$, the maker maximizes

$$U_s(t,x,p,q) = \sup_{\pi} \mathbb{E}_{t,x,p,q,s}\left[X_T + Q_T Y - \gamma \int_t^T Q_{u-}^2 \mathrm{d}\langle p\rangle_u - \gamma_T Q_T^2 p_T(1-p_T)\right]. \tag{22}$$

The running penalty prices exposure to fair-value variation, including jumps. The terminal penalty prices residual payoff variance at the horizon. This is an additive inventory-risk criterion, not a claim that the entire objective equals a mean–variance utility of terminal wealth. Execution P&L and inventory changes create additional sources of wealth variation.

Following the cash reduction in [1], cash translation invariance and conditional fair valuation give

$$\boxed{U_s(t,x,p,q) = x + qp + V_s(t,p,q).} \tag{23}$$

Indeed, $\mathbb{E}[Q_T Y \mid \mathcal{F}_T] = Q_T p_T$ for $\mathcal{F}_T$-measurable inventory, and all remaining costs depend only on (t, p, q, s) and controls. At the terminal horizon,

$$\boxed{V_0(T, p, q) = V_1(T, p, q) = -\gamma_T q^2 p(1-p).} \tag{24}$$

If a deterministic information event occurs exactly at T, apply its event condition before this post-event terminal condition.

6 Coupled Bellman equations

Under the usual dynamic-programming assumptions, the post-event value satisfies

$$\begin{aligned} 0 = {} & \partial_t V_1 + \tfrac{1}{2}\sigma^2 \partial_{pp} V_1 - \gamma q^2 \sigma^2 \\ & + \sup_{(b,a)\in\mathcal{A}(p,q)} \left\{ \mathcal{F}_{b,a}[V_1] + \lambda_u \left(\mathcal{G}_{b,a}^{K_u}[V_1] - V_1 - \gamma q^2 v_u \right) \right\}. \end{aligned} \tag{25}$$

Before the anticipated event,

$$\begin{aligned} 0 = {} & \partial_t V_0 + \tfrac{1}{2}\sigma^2 \partial_{pp} V_0 - \gamma q^2 \sigma^2 \\ & + \sup_{(b,a)\in\mathcal{A}(p,q)} \Big\{ \mathcal{F}_{b,a}[V_0] + \lambda_u \left(\mathcal{G}_{b,a}^{K_u}[V_0] - V_0 - \gamma q^2 v_u \right) \\ & \qquad + h(t) \left(\mathcal{G}_{b,a}^{K_s}[V_1] - V_0 - \gamma q^2 v_s \right) \Big\}. \end{aligned} \tag{26}$$

Arguments (t, p, q) are suppressed on the right. Both equations use (24). The pre-event equation transitions to V_1 at an anticipated event and remains in V_0 at an unscheduled shock. The system is triangular: solve V_1 first, then V_0 using the known post-event value.

The quantities σ^2, $\lambda_u v_u$, and $h v_s$ all have units of probability variance per unit time. Event penalties use q^2, not $(q')^2$, because inventory carried *into* the shock bears the jump exposure. Subsequent risk is represented by the continuation value at q'.

Crucially, $\mathcal{G}_{b,a}^K$ depends on the quotes. The maker must optimize ordinary spread revenue and stale-quote exposure jointly. A quote-independent jump operator would affect continuation values but omit this direct pickoff channel. The instantaneous risk charge $-\gamma q^2 c_s$ is control-independent at a fixed state; its effect on current quotes acts through future inventory values.

7 Economic implications and their limits

7.1 A myopic spread-widening result

To isolate stale-ask risk, omit inventory continuation values, unscheduled shocks, and other costs. Put $Z = p' - p$ and $\delta = a - p$. The one-sided reward is

$$R_h(\delta) = \Delta A e^{-k\delta}\delta - h\Delta \mathbb{E}[(Z-\delta)_+]. \tag{27}$$

At a point with no atom of Z at δ,

$$R_h'(\delta) = \Delta A e^{-k\delta}(1 - k\delta) + h\Delta \mathbb{P}(Z > \delta). \tag{28}$$

For an atomic kernel, the right derivative uses $\mathbb{P}(Z > \delta)$ and the left derivative uses $\mathbb{P}(Z \geq \delta)$; a kink optimum need not satisfy a smooth first-order condition.

Proposition 1. *Suppose admissible ask distances form* $[0, \bar{\delta}]$ *with* $\bar{\delta} > 1/k$*. If* $h > 0$ *and* $\mathbb{P}(Z > 1/k) > 0$*, every maximizer of* (27) *has* $\delta^* > 1/k$*.*

Proof. The ordinary revenue $\Delta A e^{-k\delta}\delta$ is strictly increasing for $\delta < 1/k$. The negative stop-loss term is nondecreasing in δ. Hence R_h is strictly increasing below $1/k$. At $1/k$, its right derivative is $h\Delta\mathbb{P}(Z > 1/k) > 0$, so a feasible increase improves the objective. Continuity on the compact admissible interval ensures existence of a maximizer. □

The bid-side argument uses $Z_b = p - p'$. If the no-news optimum is already at a price or displacement cap, strict widening may be infeasible. The result is a myopic statement, not a general theorem for the full inventory-dependent Bellman system. In particular, full event operators with atomic kernels can change discontinuously at fill thresholds because continuation inventory changes.

7.2 Heavy tails and withdrawal

Under a mixture (10) the ask-side jump is $Z = \rho_j(1-p)$ with probability $w_j p$ (and negative otherwise), so the myopic stale loss is

$$h\Delta\mathbb{E}[(Z-\delta)_+] = h\Delta p \sum_j w_j \big(\rho_j(1-p) - \delta\big)_+, \tag{29}$$

piecewise linear and nonincreasing in δ with kinks at the thresholds $\rho_j(1-p)$. Maximizers of (27) therefore lie at an interior stationary point of one linear piece, at a threshold, or at the endpoint.

Proposition 2. *Consider the myopic ask problem* (27) *with admissible ordinary distances* $[0, D]$*, plus the endpoint* $\delta = 1 - p > D$*, whose reward is* $\Delta A e^{-k(1-p)}(1-p) \geq 0$*.*

1. *Under a two-point kernel* K_ρ *with* $\rho(1-p) \leq D$*, the quote* $\delta = \rho(1-p)$ *has zero stale loss and positive reward, so some ordinary quote earns positive reward for every hazard* h*.*

2. *Under a mixture with a component* J *such that* $\rho_J(1-p) > D$*, every ordinary quote bears stale loss at least* $h\Delta p w_J(\rho_J(1-p) - D) > 0$*. If*

$$h\, p\, w_J(\rho_J(1-p) - D) > \frac{A}{ek}, \tag{30}$$

every ordinary quote has negative reward and the endpoint is optimal.

Proof. (1) At $\delta = \rho(1-p)$ the event never strictly crosses the quote, and $\Delta A e^{-k\delta}\delta > 0$. (2) For $\delta \leq D$, $(\rho_J(1-p) - \delta)_+ \geq \rho_J(1-p) - D$, and the remaining terms of the loss are nonnegative. Ordinary revenue $\Delta A e^{-k\delta}\delta$ is at most $\Delta A/(ek)$, attained at $\delta = 1/k$. Under (30) the loss exceeds the revenue for every $\delta \in [0, D]$, while the endpoint reward is nonnegative. □

The comparison is the substantive point. Two kernels with equal variance can imply opposite policies: the two-point kernel always offers a "safe" quote at its jump destination, which is the plateau in Figure 1A, while a kernel with a component beyond the displacement cap offers none, and a high enough hazard makes withdrawal optimal. Which case a market is in is an empirical question about the shape of its jumps, not their variance. The same caveats apply as for the first proposition: the result is myopic, and inventory continuation values can move the full policy in either direction.

7.3 Inventory management around anticipated news

The pre-event risk rate contains $\gamma q^2 h(t) v_s(p)$. Greater imminent event exposure creates an incentive to reduce absolute inventory before arrival. For a long position, the expected response is a lower bid to discourage further purchases and a lower ask relative to an inventory-neutral policy to encourage sales. Stale-quote exposure separately encourages wider quote distances.

These forces can oppose one another on a particular side. The full model does not imply that every ask rises, every spread widens, or every inventory path shrinks monotonically with hazard. Measure quote width $a-b$ and center $(a+b)/2-p$ separately, and treat the direction and magnitude of the full comparative statics as numerical results to establish.

8 An exactly timed event

If the anticipated event occurs at a deterministic $t^\star$, solve the ordinary diffusion-plus-unscheduled-news equations away from $t^\star$. The predictable event contributes an atom $v_s(p)$ to quadratic variation, and the interface condition is

$$V_0(t^{\star-}, p, q) = \sup_{(b,a)\in\mathcal{A}(p,q)} \left\{ \mathcal{G}^{K_s}_{b,a}[V_1(t^{\star+}, \cdot, \cdot)](p, q) - \gamma q^2 v_s(p) \right\}. \tag{31}$$

Here the displayed event operator evaluates only its probability and inventory arguments; the post-event time is fixed. No ordinary-flow revenue accrues over the instantaneous interface. The maker chooses pre-event quotes knowing the time but not the realized p'.

With quotes at $b = 0$ and $a = 1$, neither stale side is crossed. Such protection may amount to effective withdrawal at the event. If the research question excludes withdrawal, specify a participation constraint such as (15) and apply it consistently to all controllers. The primary experiment instead uses a finite hazard window, which preserves a tradeoff between earning ordinary-flow revenue and awaiting news.

9 Controlled experiments

9.1 Controllers and a common evaluation environment

Evaluate all policies in the same simulator, with identical actual information arrivals and stale-fill rules. The policies differ in the model they optimize. Use common random numbers where possible, without forcing identical ordinary fills when quotes imply different intensities.

Controller	Belief about event risk	Stale quotes in optimization
Diffusion baseline	Continuous fair-value risk only	Omitted
Matched-variance diffusion	Additional continuous variance with event timing	Omitted
Constant-hazard jump	Same news kernels; constant anticipated-event hazard	Included
Event-aware model	One-shot event with hazard $h(t)$	Included

For the constant-hazard comparator, preserve the one-shot state transition to isolate timing information. A recurring Poisson comparator is a separate robustness experiment because it also changes the possible event count.

9.2 Variance and event-probability matching

The anticipated event's expected quadratic variation is

$$\mathcal{V}_s = \mathbb{E}\left[\int_0^T (1 - S_{t-})h(t)v_s(p_{t-})\mathrm{d}t\right]. \tag{32}$$

At a fixed reference probability p, and assuming the event is pending at time zero with $F(0) = 0$, this reduces to

$$\int_0^T (1 - F(t))h(t)v_s(p)\mathrm{d}t = F(T)v_s(p). \tag{33}$$

Using $\int_0^T h(t)v_s(p)\mathrm{d}t$ without survival weights would count hazard after the event and generally overstate its unconditional variance contribution. When p evolves, use (32) or a simulation estimate rather than treating $v_s(p)$ as constant.

A local conditional-variance diffusion surrogate is

$$\sigma^2_{\mathrm{match,s}}(t,p) = \sigma^2(t,p) + \lambda_u(t,p)v_u(p) + (1-s)h(t)v_s(p). \tag{34}$$

Its regime transition and absorption at $0, 1$ must also be specified. Matching a local rate does not automatically match integrated expected variance when different dynamics induce different probability distributions. Verify or calibrate the latter on simulated paths.

For a constant-hazard one-shot comparator, match event probability by setting

$$\bar{h} = -\frac{\log(1 - F(T))}{T}, \qquad F(T) < 1. \tag{35}$$

With the same kernel this matches event count in expectation, but need not match realized event variance if arrival time changes the distribution of pre-event p. Report and, if needed, separately calibrate that difference.

9.3 Measurements and hypotheses

Report terminal marked wealth, the objective in (22), ordinary spread revenue, stale-fill losses, turnover, inventory at the event, time-weighted inventory exposure, and tail losses. Plot spread and quote center against both calendar time and realized event time. Include uncertainty intervals across independent simulation paths.

The principal hypothesis is that timing knowledge allows the maker to earn ordinary spread revenue while event risk is remote, then reduce exposure near the anticipated window. Whether the value of this timing information exceeds the value of matching total variance is an experimental question, not an established conclusion of the specification.

10 Numerical implementation and validation

A practical solution uses a time grid, a probability grid on $[0, 1]$, and a finite inventory grid in multiples of Δ. First solve (25) backward; then solve (26). Use monotone spatial discretization for diffusion and positive-weight interpolation for off-grid probability transitions. For K_ρ, each event expectation requires only the two destinations p^+ and p^-. Evaluate candidate quotes jointly with the event terms; include strict-crossing thresholds in the action search and document equality handling. A finite tick/action grid also makes policy attainment explicit. For a mixture kernel,

every component contributes its own pair of destinations and its own thresholds $\rho_j p$ and $\rho_j(1-p)$ to the action set; when withdrawal is admitted, add the endpoints $\delta_b = p$ and $\delta_a = 1-p$.

Choose a time-stepping method that remains stable as the hazard increases; implicit treatment or sufficiently small adaptive steps are natural options. At probability endpoints, use the reduced boundary dynamics implied by zero diffusion and degenerate kernels rather than imposing arbitrary zero values. Check convergence as time steps, probability spacing, action spacing, and inventory truncation are refined.

In simulation, an information event must execute in this order: sample p'; test the *pre-event* quotes for crossing; update cash and inventory at those quotes; change the event regime when applicable; then reprice. Repricing before the crossing test would remove the stale-execution mechanism being studied.

Useful consistency checks are:

1. Kernel normalization, mean preservation, and the variance identity $v_\rho(p) = \rho^2 p(1-p)$.
2. No second anticipated event after $S = 1$ and no scheduled-event term in the post-event Bellman equation.
3. The event mark-to-market accounting identity and nonpositive stale markouts.
4. Recovery of the diffusion-only problem when $\lambda_u = h = 0$.
5. Recovery of the quote-independent news mechanism when stale execution is disabled.
6. Agreement of simulated event frequencies and risk exposure with survival-weighted targets.

11 Monte Carlo comparison

11.1 Design and numerical solution

We compare two optimized policies in a synthetic market: a constrained Feil–Nendel diffusion benchmark and the event-aware controller. This is a comparison of specified control models, not a replication of the empirical calibration or numerical results of [1]. Both controllers use identical quote constraints, inventory limits, ordinary-flow parameters, and risk aversion. The benchmark optimizes a diffusion-only Bellman equation with running charge $\gamma q^2 \sigma^2$ and the same terminal penalty; the event-aware controller optimizes both news types, stale execution, and the full variance charge. When evaluated, both face the same actual news and pickoff mechanisms. Thus the experiment measures the combined benefit of incorporating those mechanisms; it does not isolate timing information alone.

The horizon is $T = 1$, initial cash and inventory are zero, and $p_0 = 0.5$. Trade size is $\Delta = 1$; inventories range from -5 to 5. The displacement cap is $D = 0.30$, and quotes straddle fair value. Ordinary-flow parameters are $A = 12$ and $k = 20$, with $\sigma(t,p) = 0.60p(1-p)$, $\gamma = 0.30$, and $\gamma_T = 0.02$. Unscheduled news has $\lambda_u = 0.30$ and $\rho_u = 0.15$; the anticipated event has $\rho_s = 0.50$ and hazard

$$h(t) = C \exp\left[-\frac{1}{2}\left(\frac{t-0.65}{0.07}\right)^2\right], \qquad \int_0^1 h(t)\,\mathrm{d}t = \log 10. \tag{36}$$

The event therefore occurs with probability 0.90 in continuous time. The Gaussian shape describes the *hazard*, not the event-time density; the latter also contains a survival factor. These parameters are illustrative and were fixed before inspecting the comparison's simulation results.

We solve a monotone Markov-chain approximation by backward Bellman recursion, using 81 probability nodes and 4,000 time intervals. The candidate quote-distance spacing is 0.005, augmented by exact crossing thresholds for both news kernels and by price/displacement boundaries. Inventory boundary sides are disabled for ordinary and stale execution alike. Quote optimization separates by side because a single news jump cannot cross both quotes. All reported controls are numerically optimized over this finite action set.

Diffusion uses nearest-neighbor jumps with total rate $\sigma^2/(\Delta p)^2$. News first moves to a two-point-kernel destination; positive linear interpolation is implemented in simulation as mean-preserving random rounding onto adjacent probability nodes. Crossing is tested at the unrounded destination, before repricing. The Euler transition probabilities allow at most one ordinary, diffusion, or information event per time step; the maximum total transition probability is below one. Both solvers and the simulator use this same discretization. The running penalty uses the original model variance rate, rather than the small extra variance from news rounding. Grid refinement below assesses this approximation.

The main comparison uses 30,000 independent market paths with seed 20261014. Each pair of policies shares the exogenous path, event pickoff draws, terminal Bernoulli draw $Y \sim \text{Bernoulli}(p_T)$, and uniforms for ordinary fills; fill decisions still depend on each policy's own quotes and inventory. We distinguish marked wealth $X_T + Q_T p_T$ from settled wealth $X_T + Q_T Y$. The sampled objective uses marked wealth to integrate out settlement noise, then subtracts the full true-environment running and terminal penalties for both policies. Confidence intervals below are paired normal intervals for mean differences, with standard errors computed across independent paths.

11.2 Measured performance

Metric (mean unless stated)	Diffusion benchmark	Event-aware
Marked terminal wealth ($)	0.2577	0.3571
Settled terminal wealth ($)	0.2537	0.3556
Penalized objective ($)	0.1987	0.3297
Ordinary spread revenue ($)	0.4281	0.3763
Stale-execution loss ($)	0.1684	0.0193
Running risk penalty ($)	0.0477	0.0194
Terminal risk penalty ($)	0.0113	0.0080
Ordinary fills	8.3464	7.0063
Stale fills	1.0937	0.4224
Integrated squared inventory	2.1889	1.2004
Absolute inventory before event	1.1601	0.6315
Settled wealth: 5th percentile ($)	-1.0457	-0.6763
Settled wealth: lower-tail mean ($)	-1.3786	-1.1030

Table 1: Synthetic-market results over 30,000 paired paths. Positive stale loss is a cost. The lower-tail mean averages the worst 5% of settled wealth outcomes; quantiles and tail means are descriptive estimates without confidence intervals.

The event-aware controller improves the mean penalized objective by \$0.1311, with a paired 95% confidence interval [0.1271, 0.1350]. The mean marked-wealth difference is \$0.0994 (95% interval [0.0955, 0.1033]); the mean settled-wealth difference is \$0.1019 (95% interval [0.0947, 0.1091]). Mean stale-execution losses fall by 88.6%.

At $p = 0.5$ and $q = 0$, the event-aware policy expands the spread to 0.50 near the hazard bump, placing quotes at 0.25 and 0.75. Those quotes exactly match the anticipated two-point jump

destinations, so strict crossing does not trigger a scheduled stale fill in this state. This explains the plateau in Panel A and the tradeoff between lower ordinary revenue and lower pickoff losses. Abrupt quote changes reflect the atomic news kernel and the crossing rule; they should not be interpreted as a generic smooth response to news.

These differences are conditional on this parameterization and market model. They are not evidence of superior performance in observed prediction markets. The correctly specified controller's objective advantage is expected in the ideal optimization problem; its numerical magnitude, revenue tradeoff, and out-of-objective distributional effects are the substantive measurements.

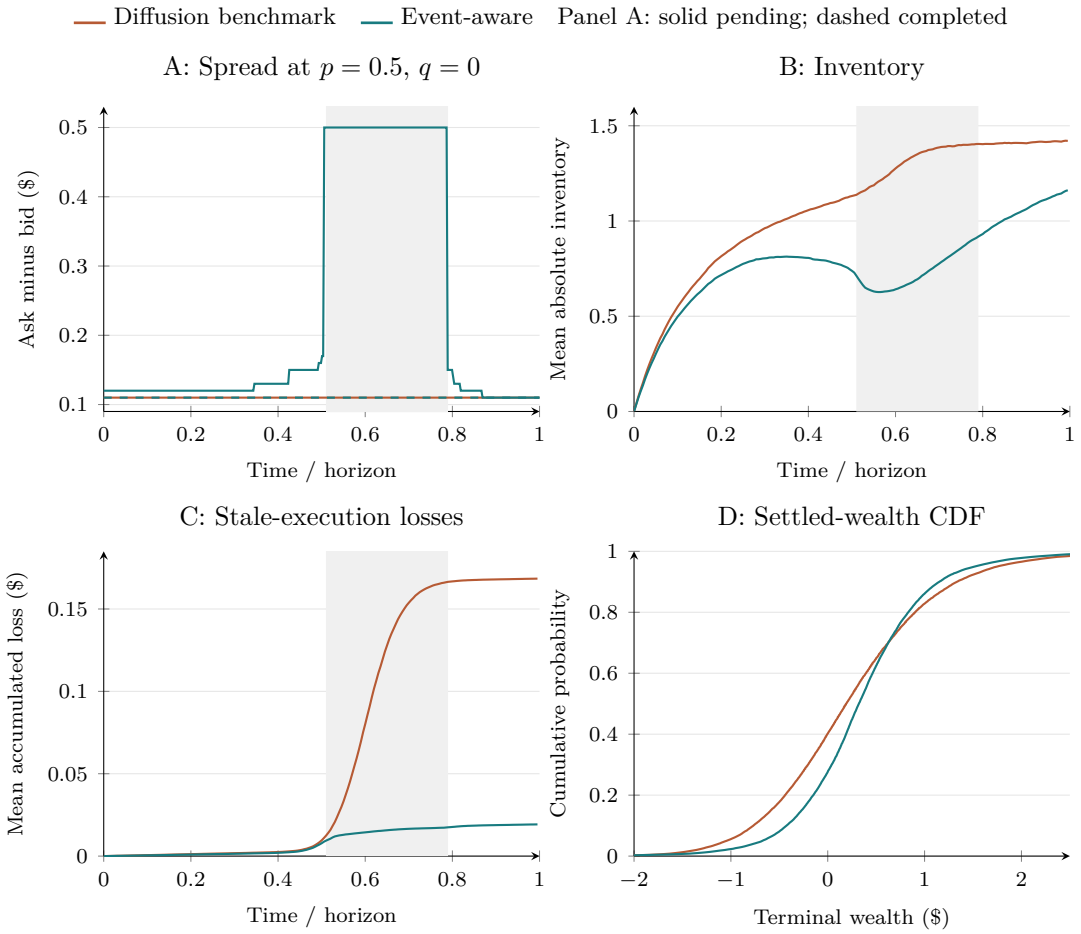


Figure 1: Optimized policies and Monte Carlo outcomes. Panel A holds probability and inventory fixed and distinguishes pending and completed event regimes; it is a policy slice, not a realized quote path. Panels B–C are unconditional sample means across shared market paths. Gray shading marks $0.65 \pm 2(0.07)$, a reference region of the hazard bump, not a confidence interval or an assertion that every event occurs there. Panel D shows empirical settled-wealth CDFs, with the horizontal range restricted for readability.

11.3 Validation, sensitivity, and limitations

The anticipated-event frequency is 0.9007, compared with the discrete-model target 0.9003; no path has more than one anticipated event. Mean unscheduled-event count is 0.3008, against target

0.30. Mean terminal probability is 0.5023 (95% interval [0.4992, 0.5053]), consistent with the initial martingale value.

Kernel means and variances, event cash/inventory accounting, and equality of both policies when news is removed are checked programmatically. We also value each stored policy directly in the true discrete environment, independently of Monte Carlo sampling. The resulting expected objectives are 0.19992 for the benchmark and 0.32911 for the event-aware controller. The corresponding Monte Carlo discrepancies are -0.51 and 0.37 standard errors.

p nodes	Time steps	Action spacing	Benchmark J	Event-aware J	Difference
41	2000	0.0100	0.19314	0.32870	0.13556
81	4000	0.0050	0.19992	0.32911	0.12920
121	8000	0.0025	0.19874	0.32939	0.13065

Table 2: Direct policy valuations in each grid's true environment. All three resolutions retain the same inventory bound. The objective advantage changes by about 1.1% from the middle to the finest grid. These are discretization checks, not Monte Carlo estimates or a proof of continuous-model convergence.

A second 30,000-path experiment uses seed 20261015, pickoff probability $\chi = 0.5$, and a reoptimized event-aware policy. The diffusion policy is unchanged. Its paired objective improvement is \$0.0570, with 95% interval [0.0531, 0.0608].

This checks sensitivity to certain pickoff, but leaves latency, informed arrival times, and time-varying ordinary flow unmodeled. Inventory limits, quote constraints, and jump sizes also merit sensitivity studies. The matched-variance and constant-hazard controllers proposed above are not part of this two-controller experiment, so these results cannot identify the pure value of knowing event time. Timing misspecification remains a further test.

The scripts `simulation/monte_carlo.py` and `simulation/report.py` reproduce the numerical study, tables, and figure. Machine-readable summaries, path-level outcomes, seeds, and grid diagnostics are retained under `results/`. All monetary amounts are dollars per unit-payoff contract; no annualization or empirical calibration is implied.

11.4 Illustrative path from the interactive prototype

To connect the aggregate experiment to individual trading decisions, Figure 2 reproduces the three charts in the supplied `index.html`: probability and quotes, inventory, and marked-to-market P&L. We execute the page's original computational functions with its default inputs and random seed 2718; we do not select a favorable path from repeated runs. Initial inventory is $q_0 = 3$, financed by $X_0 = -3(0.5)$, so initial marked wealth is zero. The timing-distribution center is 0.65, its width is 0.075, $\rho_s = 0.50$, $\lambda_u = 0.30$, and $\rho_u = 0.25$.

This prototype has 64 time intervals, 25 probability nodes on $[0.05, 0.95]$, 17 inventory states on $[-8, 8]$, and quote offsets $\{0.015, 0.030, 0.055, 0.090, 0.140\}$. Its remaining constants are $\bar{\sigma} = 0.82$, $\gamma = 0.04$, $\gamma_T = 0.01$, $A = 22$, and $k = 34$. It constructs a discrete Gaussian event-time mass with total probability 0.997 and converts that mass into conditional per-step arrival probabilities. The orange background in Panel A encodes these conditional probabilities, which are distinct from the Gaussian-shaped continuous hazard in the main Monte Carlo experiment.

The displayed path's anticipated event is $t = 0.73438$. Final marked P&L is \$1.2435 for the diffusion benchmark and \$1.2085 for the event-aware prototype. Their cumulative stale-execution losses are \$0.1315 and \$0.0215, respectively.

These are outcomes of one seeded path, not estimates of expected performance.

This illustration preserves the prototype's numerical behavior. It is a separate configuration from the validated Monte Carlo comparison above. In particular, the prototype clips probabilities to $[0.05, 0.95]$, uses nearest-node continuation values, and has different ordinary-fill approximations in its Bellman recursion and its simulator. Its figures should therefore be read as an illustration of that supplied implementation. They do not extend the main experiment's Bellman-value validation or grid-refinement conclusions. Exact replay data, default settings, and source hashes are saved in `results/prototype_path.json`; reproduction scripts are `simulation/reproduce_prototype.mjs` and `simulation/prototype_report.py`.

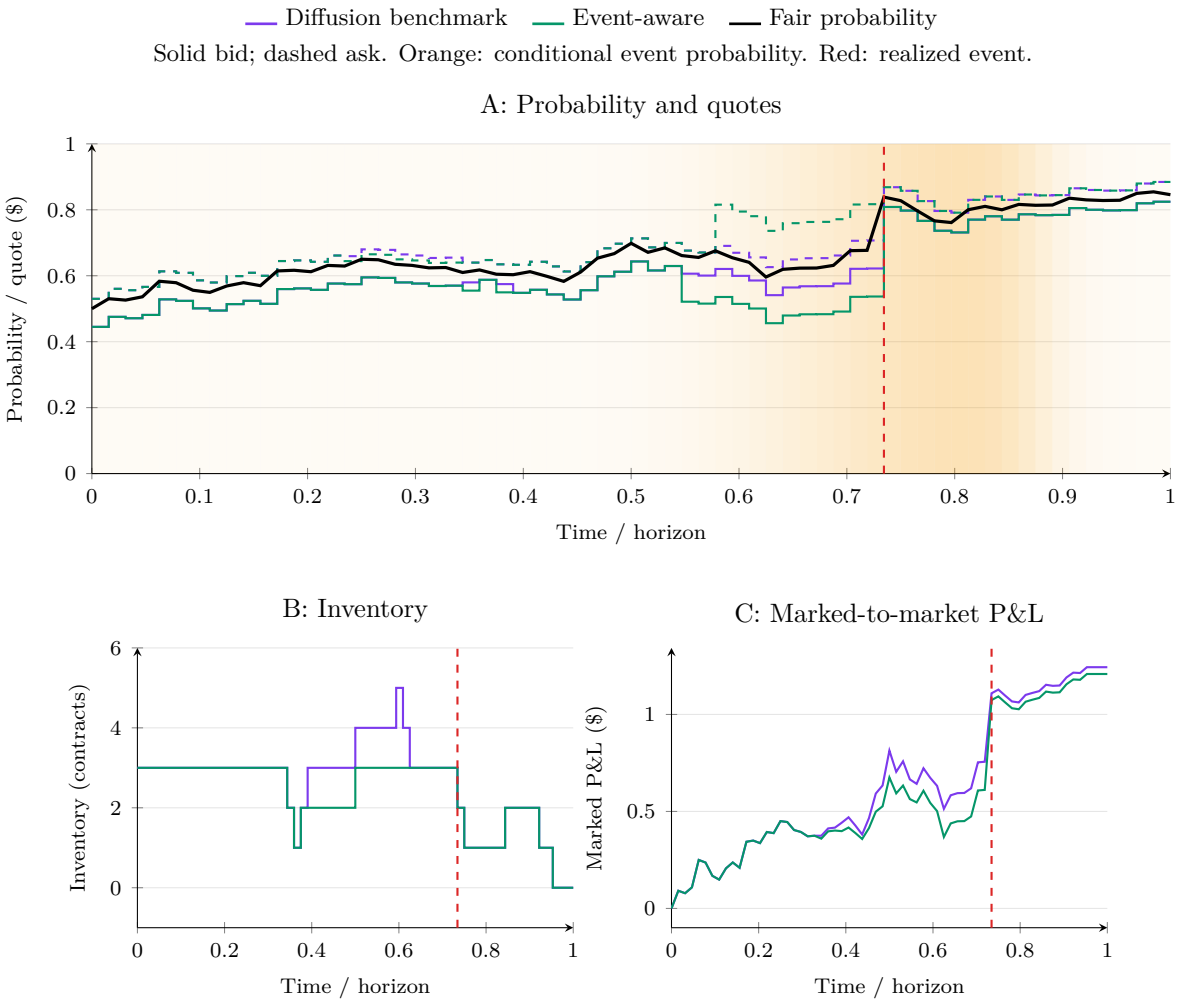


Figure 2: Reproduction of the three interactive-prototype charts at default settings and seed 2718. The upper panel shows fair probability and both policies' bids and asks; the lower panels show inventory and marked P&L. Quotes are plotted at the start of each holding interval; inventory and P&L are plotted after each transition, with their initial values included. The vertical red line marks the realized anticipated event. These physical-time conventions replace the web page's common index-based horizontal scale. The full terminal probability is included; the interactive plot truncates its last probability observation. Colors reproduce the prototype's light theme.

12 Empirical application: Kalshi daily high temperature

12.1 Market, data and replay

Kalshi's `KXHIGHNY` series lists, for each climate day, six to eight mutually exclusive contracts on the maximum temperature at Central Park. The day's information arrives on a published schedule: the station's routine hourly report is issued at minute 51 (78% of New York's reports in April–September 2026), and the National Weather Service's daily summary messages follow at fixed times. On those dates the two minutes after the report minute and after the summary minutes carry 51% of the rescaled mids' squared one-minute changes. The anticipated events are therefore known in advance; what is uncertain is how much each one moves the price.

We use an open-source replay built entirely on public data: one-minute best bid and ask from Kalshi's candlestick history, the full public trade tape, and the station's reports and summary messages. Each minute a strategy posts one 10-contract bid and ask per bin around the rescaled mid; a historical taker trade fills a quote it trades through, a quarter of it when the quote improves on the market's, and at the quote's price only if that level was then used up. Quotes are based on the previous minute's market quotes plus a 2-second delay, so they are up to about a minute stale. Positions are held to settlement, and the series charges no maker fee. Calibration uses October 2025–March 2026 (182 climate days); all results are out of sample on April–September 2026 (182 days). The same replay engine, fill rule, quote size and inventory limit (± 5 lots) are used for every strategy; only the quote construction differs.

12.2 Adapting the controller

The model of Sections 3–6 is for one contract and one anticipated event. We make three changes and no others:

1. Each bin is treated as its own binary contract with its own inventory; cross-bin settlement covariance is ignored.

2. The anticipated event recurs. The pending/completed regime is replaced by a single regime with a deterministic hazard: a Gaussian bump of width 0.6 minutes and integrated hazard 3 centered 1.5 minutes after each hourly report, and one after each summary message, each with its own kernel. Unscheduled news is set to zero; variance between reports is carried by the diffusion.

3. Ordinary fill intensities $A(p)e^{-k(p)\delta}$ are the replay's fitted fill curves by fair-value bucket, so that the execution model is shared with the strategy we compare against.

The diffusion scale is fitted on one-minute changes outside report windows, $\bar{\sigma}^2 = \sum \Delta p^2 / \sum p^2(1-p)^2$; the kernels on the change across each window, with the two-point ρ matched to its variance and the mixture weights from (13). Risk aversion γ is chosen from $\{0.1, 0.3, 1\}$ on the calibration-period replay; $\gamma_T = 0.02$, $D = 0.30$, time step 3 seconds, 101 probability nodes.

12.3 Results

The reference strategy is a closed-form Feil–Nendel quoting rule for the same series: quotes at the exponential-intensity optimum plus a fitted adverse-selection term, inventory coupled across bins through the settlement covariance $\mathrm{diag}(p) - pp^\top$ in a Riccati approximation, a hand-coded calendar that withdraws for three minutes from each report minute and 150 seconds from each summary

	hourly report	summary message
RMS move $/\sqrt{p(1-p)}$ (two-point ρ)	0.128	0.424
median $\lvert\Delta p\rvert/\sqrt{p(1-p)}$	0.017	0.09
99th percentile	0.49	1.60
share of variance in the largest 5% of moves	83%	61%
mixture weight on $\rho = 0$	0.413	0.176
$\rho \in \{0.02, 0.05, 0.1\}$	0.454	0.302
$\rho \in \{0.2, 0.35\}$	0.099	0.229
$\rho \geq 0.55$	0.034	0.294

Table 3: Report-window price moves, October 2025–March 2026, and the fitted mixture weights on $\rho \in \{0, 0.02, 0.05, 0.1, 0.2, 0.35, 0.55, 0.8, 1\}$. Diffusion scale $\bar{\sigma} = 0.070$ per $\sqrt{\text{minute}}$.

message, and a guard that posts no bids in bins the day's readings have ruled out. It is evaluated in the same replay.

Strategy	\$/day	t	days < 0	worst day	contracts/day	window fills/day
Diffusion benchmark ($\gamma = 0.3$)	3.31	3.6	40%	−35.24	1,130	153
Event-aware, two-point kernel ($\gamma = 0.3$)	5.27	6.0	32%	−40.02	1,085	112
Event-aware, mixture kernel ($\gamma = 0.1$)	10.16	10.9	16%	−25.87	1,002	34
Event-aware, mixture + withdrawal ($\gamma = 0.1$)	**11.03**	**11.7**	13%	−18.19	986	22
Diffusion benchmark + hard calendar	9.70	10.4	15%	−26.32	987	0
Reference strategy (closed form, hard calendar)	6.65	12.5	12%	−19.05	487	0
Reference strategy, no calendar or guard	1.88	3.0	36%	−34.28	586	96

Table 4: Out-of-sample replay, April–September 2026, 182 days, 10-contract quotes. t is the mean daily P&L over its standard error. "Window fills" are contracts filled inside the hourly-report windows. The hard calendar withdraws all quotes for three minutes from each report minute and 150 seconds from each summary message.

Table 4 gives three results. *(i) The ordering of Section 11 holds on real data*: the event-aware controller beats the diffusion benchmark with either kernel, by trading less inside report windows. *(ii) The kernel's shape matters more than its variance.* With the variance-matched two-point kernel the controller still takes 112 contracts a day inside the windows at −4.8 cents each: it widens to the root-mean-square jump, which is wider than needed on most reports and well inside the large ones (Table 3). The mixture kernel, which has the same order of variance but puts 3% of reports beyond $\rho = 0.55$, roughly doubles the daily P&L. *(iii) Withdrawal is learned, not imposed.* With endpoint quotes admitted, the controller withdraws its bid at 97 of 98 interior probability nodes in a report minute and at none in a quiet minute (Figure 3C), and beats the best hard-calendar policy, \$11.03 against \$9.70 a day, with a smaller worst day. This is case 2 of the proposition in Section 7.2.

12.4 Between reports

Outside the report windows the event-aware and diffusion controllers quote almost identically, so the gap between our controllers and the reference strategy in Table 4 (\$9.70 against \$6.65 with the same calendar) is a difference in ordinary quoting, not in event handling. Their quote adds a fitted adverse-selection term to the exponential-intensity optimum and couples bins through the settlement covariance. Removing either does not close the gap: with the adverse term at zero they

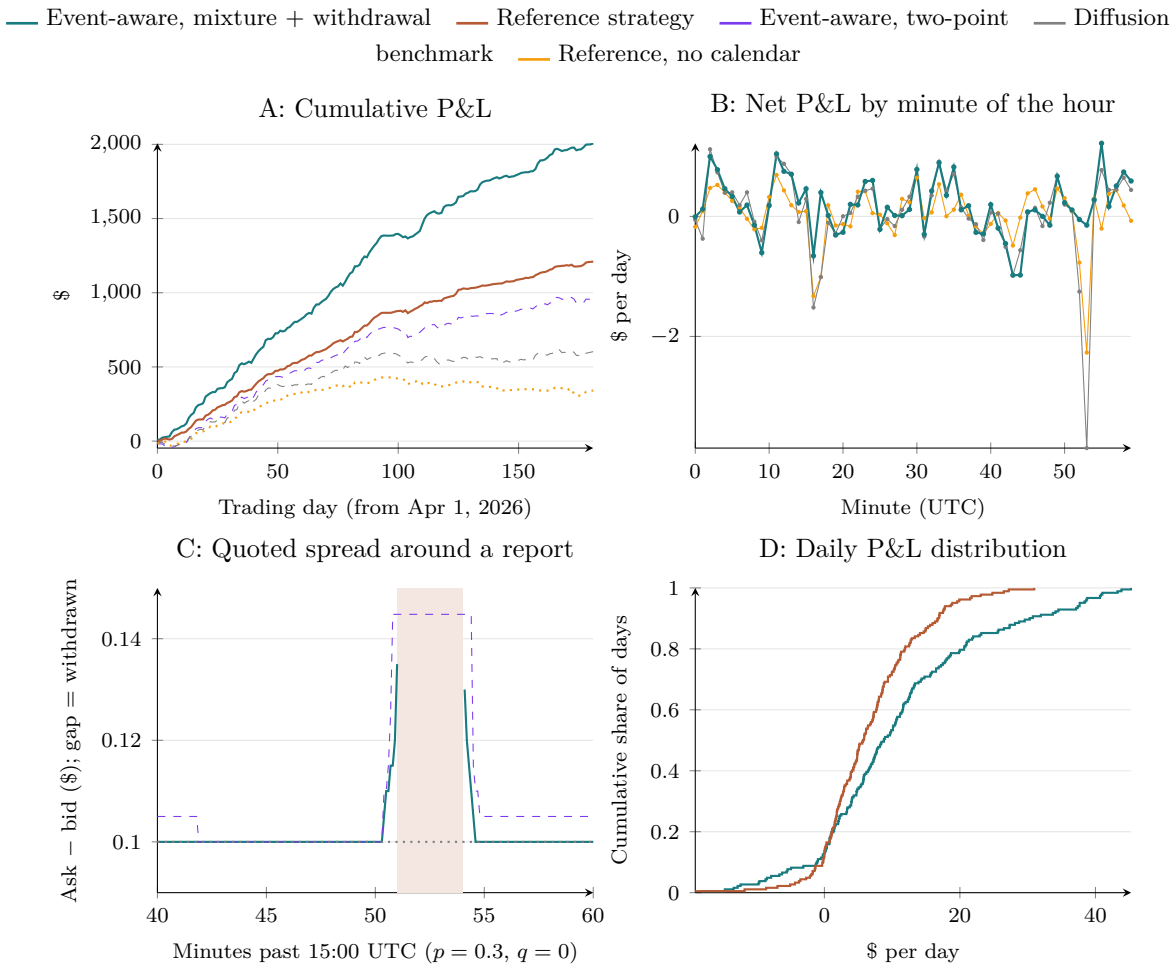


Figure 3: Out-of-sample replay, April–September 2026. Panel B sums each fill's settlement P&L by the minute of the hour in which it was quoted; the hourly report is issued at minute 51. Panel C is a policy slice at fixed probability and inventory; the shaded band is the published strategy's hard calendar, during which it posts nothing, and gaps in the event-aware line are withdrawal.

trade as much as we do (919 against 979 contracts a day) but earn $6.35 a day; without the cross-bin terms, $6.85. At comparable volume our fills capture about 0.45 cents more edge per contract. Candidate explanations, not yet separated, are price rounding (we always round away from fair value), the inventory penalty (our γ is chosen on the calibration period), and their adjustment of the fill rate for adverse selection.

12.5 Limitations

The replay has no historical order books. It cannot see the queue ahead of a quote at its price, other makers' reaction to our quotes, or depth beyond the trade tape, and quotes are formed from one-minute data. These limits apply equally to every row of Table 4, so the comparisons between controllers are better supported than the absolute dollar figures. Capacity is small: 10-contract quotes take under 1% of the series' volume. One series and one six-month test period are reported; other cities have different stations, report minutes and participants. The hazard bumps are placed from the published schedule, not estimated; their width and mass are not tuned. A live test with recorded books is the next step.

A Optional partial-pickoff extension

The baseline sets pickoff probability to one. For $0 \le \chi \le 1$, an independent Bernoulli pickoff decision conditional on a crossing gives

$$\mathcal{G}^{K,\chi}_{b,a}[W] = \chi \mathcal{G}^{K}_{b,a}[W] + (1-\chi) \int W(t, p', q) K(\mathrm{d}p' \mid p). \tag{37}$$

This mixes continuation inventories as well as markout losses. Merely multiplying the loss term by χ while always changing inventory would be inconsistent. The probability process and its risk charge are unchanged. Certain pickoff is a maximal execution-probability case; it is not automatically a bound on optimized total value because inventory effects also change.

B Specification status and provenance

The shared discussion is the immediate source for this specification; Section 2 identifies its published foundations and empirical motivation. The event kernels, stale-event operator, coupled Bellman equations, deterministic event interface, and experimental comparison are presented here as the proposed framework. This write-up supplies explicit admissibility and timing conventions, qualifies the spread result, and corrects the unconditional variance-matching expression for one-shot survival. Numerical parameters, participation limits, inventory limits, and grid resolution are explicitly fixed for the synthetic experiment in Section 11. Its results are numerical model comparisons, without empirical calibration. Bibliographic details and the limited empirical claims in Section 2 were checked against the cited papers or their publisher/institutional records. The bibliography distinguishes the direct foundation from contextual work; it is not an exhaustive literature review.

References

[1] Dominik Feil and Max Nendel. *Optimal Market Making in Prediction Markets*. arXiv preprint arXiv:2607.17991, 2026. https://arxiv.org/abs/2607.17991.

[2] Marco Avellaneda and Sasha Stoikov. High-frequency trading in a limit order book. *Quantitative Finance*, 8(3):217–224, 2008. https://doi.org/10.1080/14697680701381228.

[3] Álvaro Cartea, Sebastian Jaimungal, and José Penalva. *Algorithmic and High-Frequency Trading*. Cambridge University Press, 2015. ISBN 978-1-107-09114-6. Publisher front matter and contents.

[4] Alex Frino and Amelia Hill. Intraday futures market behaviour around major scheduled macroeconomic announcements: Australian evidence. *Journal of Banking & Finance*, 25(7):1319–1337, 2001. https://doi.org/10.1016/S0378-4266(00)00132-1.

[5] Milla Siikanen, Juho Kanniainen, and Jaakko Valli. Limit order books and liquidity around scheduled and non-scheduled announcements: Empirical evidence from NASDAQ Nordic. *Finance Research Letters*, 21:264–271, 2017. https://doi.org/10.1016/j.frl.2016.12.016.